

Variational autoencoders (VAE) is a type of probability generative model.

Its core idea is to model the "true" distribution as

$$p(x) = \int \underbrace{p(z)}_{\text{Latent Variable}} \underbrace{p(x|z)}_{\text{Conditional Likelihood given Latent Variable}} dz$$

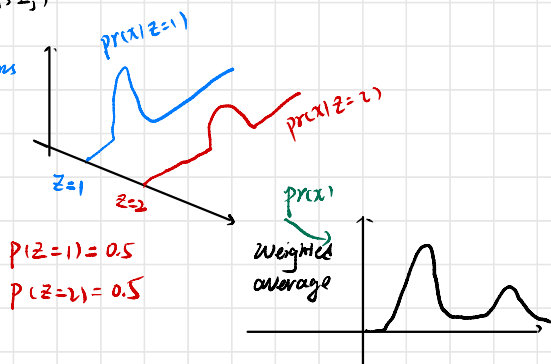
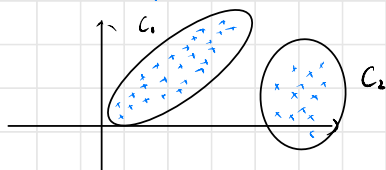
Examples:

1. Gaussian Mixture Model

Latent variable: $z_{ij} = \begin{cases} 1 & \text{if } x_i \in C_j \\ 0 & \text{otherwise} \end{cases}$

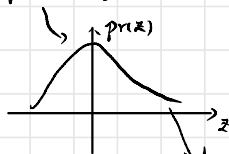
Conditional Likelihood: $p(x_i | z_{ij}=1) = \mathcal{N}(x_i; \mu_j, \Sigma_j)$

both are simple, but can model complex distributions

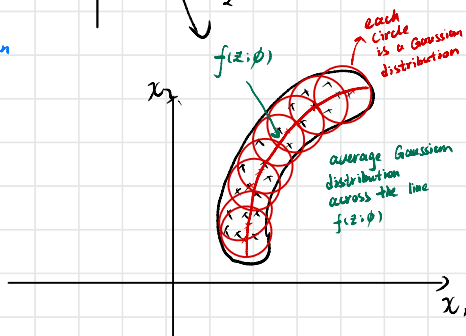
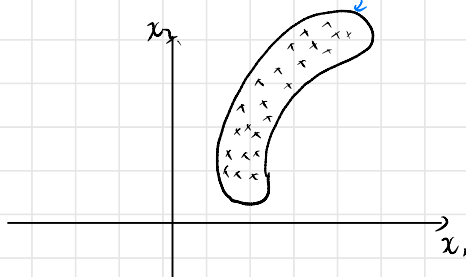


2. 2D non-linear latent variable model

$$\int p(z) \mathcal{N}(f(z; \phi), \Sigma) dz$$



How to model this distribution



Conclusion for both: Weighted average of simple Gaussian Distributions can be used to approximate very complicated distribution.

The problem of VAE:

$$p(x) = \int p(x) p(x|z; \phi) dz$$

$$= \int \underbrace{N(z; 0, I) N(x; f(z; \phi), \sigma^2 I)}_{\text{has no analytical form}} dz$$

↙
We can approximate this as

↘ has no analytical form

$E_{z \sim N(0, I)} (N(x; f(z; \phi), \sigma^2 I))$ to estimate the marginal likelihood of x

However, because dimension of z is high, we need very large sample size to make this work.

(For a given image point x , if we sample randomly from $z \sim N(0, I)$, then it is very likely (>99.9%) that $N(x; f(z; \phi), \sigma^2 I) \approx 0$)

So it is hard to maximize the marginal likelihood $p(x)$ directly.

Instead, we will try to push up the lower bound of the marginal likelihood.

$$\log p(x|\phi) = \log \left(\int p(x, z|\phi) dz \right)$$

$$= \log \left(\int q(z) \frac{p(x, z|\phi)}{q(z)} dz \right)$$

$$\geq \int \underbrace{q(z) \log \frac{p(x, z|\phi)}{q(z)}}_{\text{this is known as evidence lower bound (elbo)}} dz$$

Jensen's inequality

↘ this is known as evidence lower bound (elbo)

↙ this proposed $q(z)$ can be anything, and it will decide how for elbo is from $p(x|\phi)$. We can learn it as well, and we denote it as $q(z|\theta)$ below.

(as long as it is a valid distribution...)

change θ : move elbo → target
change ϕ : push up elbo itself

$$\text{ELBO}(\theta, \phi) = \int \underbrace{q(z|\theta) \log \frac{p(x, z|\phi)}{q(z|\theta)}}_{\text{evidence lower bound (elbo)}} dz$$

another perspective of VAE: the neural network architecture to maximize ELBO is VAE.

Analyze ELBO 1: Tightness of the bound (when $ELBO(\theta, \phi) = \log \text{pr}(x|\theta)$)

$$\begin{aligned}
 ELBO(\theta, \phi) &= \int q(z|\theta) \log \frac{\text{pr}(x, z|\phi)}{q(z|\theta)} dz \\
 &= \int q(z|\theta) \log \frac{\text{pr}(z|x, \phi) \text{pr}(x|\phi)}{q(z|\theta)} dz \\
 &= \int q(z|\theta) \log \text{pr}(x|\phi) dz - \int q(z|\theta) \log \frac{q(z|\theta)}{\text{pr}(z|x, \phi)} dz \\
 &= \log \text{pr}(x|\phi) - \underbrace{KL(q(z|\theta), \text{pr}(z|x, \phi))}
 \end{aligned}$$

→ when the proposed distribution $q(z|\theta) = \text{pr}(z|x, \phi)$, we have
 $ELBO(\theta, \phi) = \log \text{pr}(x|\phi)$
 However, $\text{pr}(z|x, \phi) = \frac{\text{pr}(z) \text{pr}(x|z, \phi)}{\text{pr}(x|\phi)}$
 could be very complex.

Analyze of ELBO 2: Construct loss function from ELBO

$$\begin{aligned}
 ELBO(\theta, \phi) &= \int q(z|\theta) \log \frac{\text{pr}(x, z|\phi)}{q(z|\theta)} dz \\
 &= \int q(z|\theta) \log \frac{\text{pr}(x|z, \phi) \text{pr}(z)}{q(z|\theta)} dz \quad \text{prior } z \sim N(0, I) \\
 &= \int q(z|\theta) \log \text{pr}(x|z, \phi) dz - \int q(z|\theta) \log \frac{q(z|\theta)}{\text{pr}(z)} dz \\
 &= \underbrace{\int q(z|\theta) \log \text{pr}(x|z, \phi) dz}_{\text{reconstruction error}} - \underbrace{KL(q(z|\theta), \text{pr}(z))}_{\text{the proposed distribution } q(z|\theta) \text{ will be penalized if deviates from prior } \text{pr}(z)}
 \end{aligned}$$

$$= \mathbb{E}_{z \sim q(z|\theta)} (\log \text{pr}(x|z, \phi))$$

Combining the above two analyses for ELBO, we now know that

1. We want the proposed distribution $q(z|\theta)$ to be close to $\text{pr}(z|x, \phi)$, so that $\text{ELBO} \rightarrow \text{target}$.
2. We want the proposed distribution to minimize reconstruction error, (which means it should provide enough information about the observation x)
3. We also want $q(z|\theta)$ to be close to $\text{pr}(z)$, so it should stay simple.

Solution: we let

$$q(z|\theta) = q(z|x, \theta) = \mathcal{N}(z; g_\mu(z, \theta), g_\sigma(z, \theta))$$

↳ Variational approximation to approximate $\text{pr}(z|x, \phi)$

↳ this could be very complex. So this approximation may not be good. $\Rightarrow$ ELBO is not close to log marginal likelihood.

Summary

Loss function: $\mathcal{L}(\theta, \phi) = \int q(z|\theta) \log \text{pr}(x|z, \phi) dz + \text{KL}(q(z|\theta), \text{pr}(z))$

Here z is also high dimension.

So why is this approximation still work, but

$$\int \text{pr}(z) \text{pr}(x|z, \phi) dz = \mathbb{E}_{z \sim \mathcal{N}(0, I)} (\text{pr}(x|z, \phi))$$

$$= - \int q(z|x, \theta) \log \text{pr}(x|z, \phi) dz + \text{KL}(q(z|x, \theta), \text{pr}(z))$$

$$= - \mathbb{E}_{z \sim q(z|x, \theta)} (\log \text{pr}(x|z, \phi)) dz + \text{KL}(q(z|x, \theta), \text{pr}(z))$$

↳ since $q(z|x, \theta)$ is Gaussian, has explicit form

$$\approx \frac{1}{N} \sum_{i=1}^N \text{pr}(x|z_i, \phi) \ll \approx - \frac{1}{N} \sum_{i=1}^N \log \text{pr}(x|z_i, \phi) dz + \text{KL}(q(z|x, \theta), \text{pr}(z))$$

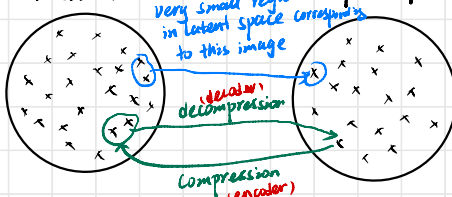
will fail?

Reason: if $z \sim \mathcal{N}(0, I)$, then $\text{pr}(x|z_i, \phi) \approx 0$.

if however $z \sim q(z|x, \theta)$, then $\text{pr}(x|z_i, \phi) > 0$

Latent Space $\mathcal{N}(0, I)$

Sample Space



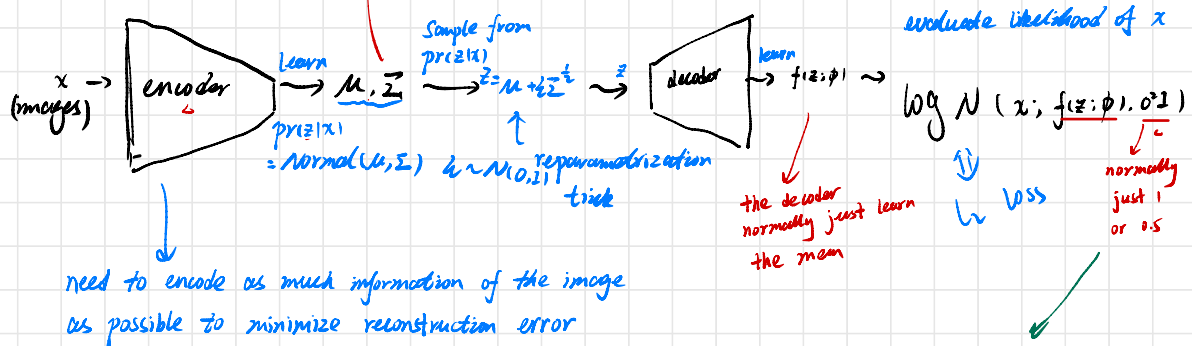
The compression step $q(\cdot|x, \theta)$ narrows down the scope in latent space to a small region.

if we let this become $2\text{KL}(q(z|x, \theta), \text{pr}(z))$
 $\downarrow$
 $2\uparrow$, encourage the features to be disentangled.

This Σ need to be Symmetric and Semi-positive definite.

1. normally, we assume $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2)$
2. Otherwise, we use Cholesky decomposition

$\Sigma = LL^T$, where L is lower-triangular matrix with positive values on its diagonal



need to encode as much information of the image as possible to minimize reconstruction error

VS

need to be simple enough so that $KL(q(z|\theta), N(0, I))$ is small

Note: 1. Remember the target is to maximize $\log p(x|\phi)$. The loss function of VAE is maximizing ELBO directly. So how about the distance between ELBO and $\log p(x|\phi)$?

$$\log p(x|\phi) = \text{ELBO}(\theta, \phi) + KL(q(z|\theta), p(z|x, \phi))$$

the marginal likelihood depends on decoder only. So for a given decoder ϕ , it is a constant for encoder

both encoder & decoder causes about ELBO only

as along as encoder is pushing up ELBO, it is minimizing the distance with $p(z|x, \phi)$ although we don't know what is $p(z|x, \phi)$

2. To generate image from VAE,

$z \sim \mathcal{N}(0, I) \rightarrow \text{decoder} \rightarrow \text{generated image}$

3. In order to improve the quality of the generated images from VAE, it seems not enough to model prior $p(z)$ and latent space $p(z|x)$ as Gaussian. Diffusion can generate higher quality images and can be consider VAE with hierarchical priors.

evaluate likelihood of x

$$\log N(x; f(z; \phi), 0.1)$$

loss

normally just 1 or 0.5

the decoder normally just learn the mean

why does the decoder not learn 0?

1. if image is hard, it could push $0 \rightarrow \infty$
2. if image is easy, it would push $0 \rightarrow 0$

Because the marginal likelihood does not depend only on decoder, we can also think the encoder here is only trying to help the decoder to learn.

without encoder: decoder needs to evaluate the whole latent space to evaluate $p(x|\phi)$

with encoder: encoder helps to narrow down the space to $q(z|x, \theta)$